TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	81.7 ± 2.37
Correlation Stability	79.2 ± 3.31
Distribution Consistency	87.6 ± 1.56
Entropy Preservation	91.6 ± 1.36
Mean Z Score Drift	0.1 ± 0.03
Categorical Fidelity	90.3 ± 2.69
Privacy Risk	50.2 ± 1.17
Data Density	100.0 ± 0.01
Coverage Score	16.6 ± 3.17
Feature Accuracy	84.1 ± 3.11
Novelty Score	0.4 ± 0.01

TabuLLM: AI Benchmark Report

Dataset Audit: indian_liver_patient_dataset.csv

1. Dataset Portrait

Attribute	Value
Rows	583
Columns	11
Numeric features	10
Categorical features	1
Missing cells (%)	0.1%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	gpt-4.1
Folds (*13)	10
Context % (*14)	20%
Generation % (*15)	100%
Join context (*16)	True

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
Age	44.746	44.313	-0.433	1.0%	OK
TB	3.299	2.184	-1.115	33.8%	Drift
DB	1.486	0.892	-0.594	40.0%	Drift
Alkphos	290.576	264.936	-25.641	8.8%	OK
Sgpt	80.714	58.633	-22.081	27.4%	Drift
Sgot	109.911	79.738	-30.173	27.5%	Drift
TP	6.483	6.682	+0.198	3.1%	OK
ALB	3.142	3.271	+0.130	4.1%	OK
A/G Ratio	0.947	0.967	+0.020	2.1%	OK
Selector	1.286	1.412	+0.126	9.8%	OK

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	81.7 ± 2.37
Correlation Stability (*2)	79.2 ± 3.31
Distribution Consistency (*3)	87.6 ± 1.56
Entropy Preservation (*4)	91.6 ± 1.36
Mean Z Score Drift (*5)	0.1 ± 0.03
Categorical Fidelity (*6)	90.3 ± 2.69
Privacy Risk (*7)	50.2 ± 1.17
Data Density (*8)	100.0 ± 0.01
Coverage Score (*9)	16.6 ± 3.17

Feature Accuracy (*10)	84.1 ± 3.11
Novelty Score (*11)	0.4 ± 0.01

*[1] (*12) Algorithm: Contextual constraint or model parameter.*

*[2] (*13) Folds: Contextual constraint or model parameter.*

*[3] (*14) Context %: Contextual constraint or model parameter.*

*[4] (*15) Generation %: Contextual constraint or model parameter.*

*[5] (*16) Join context: Contextual constraint or model parameter.*

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6863247863247863	0.6589743589743591	0.6760683760683761	-0.010256410256410243	
F1	0.7971299319768493	0.7407567100511275	0.7721831169479974	-0.024946815028852088	

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.653846153846154	0.6034188034188035	0.6777777777777778	0.02393162393162393	
F1	0.7574969615135021	0.6922120171434909	0.7717567806043792	0.01425981909087709	

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6846153846153846	0.6376068376068377	0.7008547008547008	0.016239316239316227	
F1	0.789990711889234	0.7255648038625754	0.791949677014798	0.0019589651255641272	

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7068376068376068	0.6547008547008546	0.6820512820512821	-0.02478632478632479	
F1	0.8279091624529537	0.7172601244098402	0.7728824411889545	-0.05502672126399909	*

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6914529914529914	0.6623931623931625	0.6829059829059829	-0.00854700854700854	
F1	0.7996066996360383	0.7490485485082722	0.7815751231267075	-0.018031576509330703	

Model: LR

TabuLLM: AI Benchmark Report

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7034188034188034	0.6427350427350427	0.6914529914529914	-0.011965811965811956	
F1	0.8108734553482932	0.7246542994573418	0.7833259822993468	-0.027547473048946447	

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7025641025641025	0.6598290598290599	0.7	-0.002564102564102555	
F1	0.7997903571549948	0.7469895906431198	0.7935582418920631	-0.006232115262931781	

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7051282051282051	0.652991452991453	0.7051282051282052	1.1102230246251566e-17	
F1	0.8103793645297219	0.7402389279251608	0.7997279725661093	-0.010651391963612622	

Model: AB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6743589743589744	0.6641025641025642	0.6863247863247863	0.011965811965811979	
F1	0.7821343746683586	0.7463810230225055	0.7778261239757981	-0.004308250692560567	

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.558974358974359	0.6632478632478633	0.6076923076923076	0.0487179487179487	*
F1	0.5587432119380917	0.7339407291665083	0.6377807082987751	0.07903749636068345	*

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	65	Female	0.7	0.1	187	16	18	6.8	3.3	0.9
2	62	Male	10.9	5.5	699	64	100	7.5	3.2	0.74
3	62	Male	7.3	4.1	490	60	68	7.0	3.3	0.89
4	58	Male	1.0	0.4	182	14	20	6.8	3.4	1.0
5	72	Male	3.9	2.0	195	27	59	7.3	2.4	0.4
6	46	Male	1.8	0.7	208	19	14	7.6	4.4	1.3
7	26	Female	0.9	0.2	154	16	12	7.0	3.5	1.0
8	29	Female	0.9	0.3	202	14	11	6.7	3.6	1.1
9	17	Male	0.9	0.3	202	22	19	7.4	4.1	1.2
10	55	Male	0.7	0.2	290	53	58	6.8	3.4	1.0
11	57	Male	0.6	0.1	210	51	59	5.9	2.7	0.8
12	72	Male	2.7	1.3	260	31	56	7.4	3.0	0.6
13	64	Male	0.9	0.3	310	61	58	7.0	3.4	0.9
14	74	Female	1.1	0.4	214	22	30	8.1	4.1	1.0
15	61	Male	0.7	0.2	145	53	41	5.8	2.7	0.87
16	25	Male	0.6	0.1	183	91	53	5.5	2.3	0.7
17	38	Male	1.8	0.8	342	168	441	7.6	4.4	1.3
18	33	Male	1.6	0.5	165	15	23	7.3	3.5	0.92
19	40	Female	0.9	0.3	293	232	245	6.8	3.1	0.8
20	40	Female	0.9	0.3	293	232	245	6.8	3.1	0.8

TabuLLM: AI Benchmark Report

B.1 Audit Trace: Fold 1 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	70	Male	1.3	0.3	690	93	40	3.6	2.7	0.7
2	28	Female	1.0	0.3	90	18	108	6.8	3.1	0.8
3	47	Male	0.9	0.2	192	38	24	7.3	4.3	1.4
4	42	Male	1.0	0.3	154	38	21	6.8	3.9	1.3
5	60	Male	0.7	0.2	171	31	26	7.0	3.5	1.0
6	50	Female	1.0	0.3	191	22	31	7.8	4.0	1.0
7	54	Male	1.2	0.4	210	28	35	6.7	3.5	1.0
8	36	Female	0.9	0.2	175	22	25	7.1	3.8	1.1
9	45	Male	2.5	1.1	320	65	98	5.9	2.6	0.7
10	29	Male	0.8	0.2	140	19	22	7.4	4.0	1.2
11	62	Male	6.2	2.9	420	88	130	5.2	2.0	0.6
12	40	Female	1.1	0.3	200	30	40	6.8	3.2	0.9
13	33	Male	7.1	3.7	196	622	497	6.9	3.6	1.09
14	38	Male	2.2	1.0	310	119	42	7.9	4.1	1.0
15	64	Male	3.0	1.4	248	46	40	6.5	3.2	0.9
16	40	Female	0.9	0.2	285	32	27	7.7	3.5	0.8

B.2 Audit Trace: Fold 2 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	29	Female	0.7	0.1	162	52	41	5.2	2.5	0.9
2	62	Male	7.3	4.1	490	60	68	7.0	3.3	0.89
3	58	Female	2.8	1.3	670	48	79	4.7	1.6	0.5
4	51	Male	0.8	0.2	367	42	18	5.2	2.0	0.6
5	35	Male	2.0	1.1	226	33	135	6.0	2.7	0.8
6	46	Female	1.4	0.4	298	509	623	3.6	1.0	0.3
7	53	Male	2.3	1.1	312	48	78	6.7	3.2	0.9
8	41	Female	0.9	0.2	205	32	40	7.2	3.6	1.0
9	60	Male	5.6	2.9	420	110	120	6.3	2.9	0.8
10	27	Female	0.7	0.2	140	25	28	6.5	3.1	1.1
11	35	Male	1.6	0.6	210	38	55	7.0	3.5	1.2
12	58	Female	3.2	1.4	375	62	90	5.9	2.7	0.7
13	50	Female	1.0	0.3	191	22	31	7.8	4.0	1.0
14	52	Female	0.6	0.1	194	10	12	6.9	3.3	0.9
15	30	Male	0.8	0.2	182	46	57	7.8	4.3	1.2
16	58	Male	0.8	0.2	298	33	59	6.2	3.1	1.0

B.3 Audit Trace: Fold 3 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	33	Male	2.0	1.0	258	194	152	5.4	3.0	1.25
2	21	Male	0.7	0.2	135	27	26	6.4	3.3	1.0
3	74	Male	1.0	0.3	175	30	32	6.4	3.4	1.1
4	55	Male	0.9	0.2	116	36	16	6.2	3.2	1.0
5	64	Male	1.1	0.4	201	18	19	6.9	4.1	1.4
6	34	Male	4.1	2.0	289	875	731	5.0	2.7	1.1
7	47	Male	1.3	0.5	205	38	45	7.2	3.6	1.1
8	53	Female	2.1	1.0	312	56	68	6.8	3.2	0.9
9	28	Male	0.8	0.2	178	29	32	7.0	3.8	1.3
10	61	Male	4.5	2.2	480	120	210	6.2	2.8	0.8
11	36	Female	0.7	0.2	150	22	27	7.5	4.0	1.2
12	44	Male	3.2	1.4	265	62	80	6.5	2.9	0.7
13	44	Female	1.9	0.6	298	378	602	6.6	3.3	1.0
14	30	Male	0.8	0.2	182	46	57	7.8	4.3	1.2
15	53	Male	0.9	0.4	238	17	14	6.6	2.9	0.8
16	28	Male	0.8	0.3	190	20	14	4.1	2.4	1.4

B.4 Audit Trace: Fold 4 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	34	Female	0.8	0.2	192	15	12	8.6	4.7	1.2
2	35	Male	0.9	0.2	190	25	20	6.4	3.6	1.2
3	46	Male	1.8	0.7	208	19	14	7.6	4.4	1.3
4	27	Male	1.2	0.4	179	63	39	6.1	3.3	1.1
5	14	Male	1.4	0.5	269	58	45	6.7	3.9	1.4
6	60	Male	2.1	1.0	191	114	247	4.0	1.6	0.6
7	43	Male	1.2	0.5	218	36	40	6.7	3.1	0.9
8	29	Female	0.7	0.2	175	22	18	7.2	4.0	1.3
9	54	Male	2.0	0.9	410	92	110	6.1	2.7	0.7
10	37	Male	1.5	0.6	195	44	29	7.0	3.5	1.0
11	61	Female	0.9	0.3	142	17	20	5.9	2.8	0.8
12	48	Male	3.3	1.7	350	68	73	6.5	2.9	0.7
13	42	Male	0.7	0.2	197	64	33	5.8	2.4	0.7
14	69	Male	0.9	0.2	215	32	24	6.9	3.0	0.7
15	53	Male	1.6	0.9	178	44	59	6.5	3.9	1.5
16	49	Male	1.3	0.4	206	30	25	6.0	3.1	1.06

B.5 Audit Trace: Fold 5 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	38	Male	3.1	1.6	253	80	406	6.8	3.9	1.3
2	26	Male	0.6	0.1	110	15	20	2.8	1.6	1.3
3	32	Male	32.6	14.1	219	95	235	5.8	3.1	1.1
4	24	Male	1.0	0.2	189	52	31	8.0	4.8	1.5
5	48	Male	0.7	0.2	326	29	17	8.7	5.5	1.7
6	58	Male	0.8	0.2	123	56	48	6.0	3.0	1.0
7	44	Male	2.1	1.0	210	38	65	6.7	3.2	0.9
8	27	Female	0.8	0.2	170	19	22	7.2	4.0	1.2
9	53	Male	1.6	0.7	245	55	120	5.9	2.7	0.8
10	36	Male	3.5	1.8	320	72	210	6.2	3.1	1.0
11	61	Female	0.7	0.1	160	24	19	6.8	3.5	1.1
12	49	Male	1.3	0.4	190	41	58	7.1	3.8	1.3
13	33	Male	2.1	0.7	205	50	38	6.8	3.0	0.7
14	22	Female	6.7	3.2	850	154	248	6.2	2.8	0.8
15	48	Male	5.8	2.5	802	133	88	6.0	2.8	0.8
16	84	Female	0.7	0.2	188	13	21	6.0	3.2	1.1

B.6 Audit Trace: Fold 6 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	42	Male	0.8	0.2	114	21	23	7.0	3.0	0.7
2	55	Male	1.8	9.0	272	22	79	6.1	2.7	0.7
3	33	Male	0.7	0.1	168	35	33	7.0	3.7	1.1
4	51	Male	0.8	0.2	175	48	22	8.1	4.6	1.3
5	21	Male	0.8	0.2	183	33	57	6.8	3.5	1.0
6	72	Male	1.7	0.8	200	28	37	6.2	3.0	0.93
7	44	Male	1.1	0.3	210	32	45	6.7	3.2	0.9
8	58	Female	0.8	0.2	172	28	24	7.5	3.8	1.1
9	36	Male	2.3	1.0	295	102	140	6.2	2.9	0.8
10	27	Female	0.7	0.1	160	20	18	7.1	3.7	1.0
11	61	Male	1.6	0.6	225	34	52	6.0	2.8	0.7
12	49	Male	3.5	1.7	410	78	110	5.8	2.5	0.6
13	50	Male	0.9	0.3	194	190	73	7.5	3.9	1.0
14	49	Male	0.6	0.1	218	50	53	5.0	2.4	0.9
15	18	Male	0.8	0.2	228	55	54	6.9	4.0	1.3
16	30	Female	0.7	0.2	194	32	36	7.5	3.6	0.92

B.7 Audit Trace: Fold 7 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	50	Male	0.9	0.2	202	20	26	7.2	4.5	1.66
2	35	Male	26.3	12.1	108	168	630	9.2	2.0	0.3
3	56	Male	1.0	0.3	195	22	28	5.8	2.6	0.8
4	46	Male	10.2	4.2	232	58	140	7.0	2.7	0.6
5	72	Male	0.7	0.2	185	16	22	7.3	3.7	1.0
6	58	Male	0.8	0.2	180	32	25	8.2	4.4	1.1
7	41	Male	1.2	0.4	210	28	36	6.7	3.5	1.1
8	59	Male	2.5	1.0	480	110	160	7.0	2.8	0.7
9	35	Female	0.8	0.2	175	22	29	7.5	4.0	1.2
10	62	Male	1.6	0.7	320	44	57	6.2	3.0	0.9
11	27	Male	0.9	0.3	188	35	41	7.8	4.2	1.2
12	48	Female	1.3	0.5	260	38	52	6.3	2.7	0.8
13	13	Female	0.7	0.2	350	17	24	7.4	4.0	1.1
14	54	Female	23.2	12.6	574	43	47	7.2	3.5	0.9
15	63	Male	0.9	0.2	194	52	45	6.0	3.9	1.85
16	46	Male	0.6	0.2	290	26	21	6.0	3.0	1.0

B.8 Audit Trace: Fold 8 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	30	Female	0.8	0.2	158	25	22	7.9	4.5	1.3
2	53	Male	1.6	0.9	178	44	59	6.5	3.9	1.5
3	10	Female	0.8	0.1	395	25	75	7.6	3.6	0.9
4	40	Male	0.7	0.2	176	28	43	5.3	2.4	0.8
5	48	Male	0.7	0.2	326	29	17	8.7	5.5	1.7
6	50	Male	7.3	3.7	92	44	236	6.8	1.6	0.3
7	52	Male	2.4	1.1	298	65	78	6.9	3.2	0.8
8	36	Female	0.8	0.2	172	30	27	8.1	4.4	1.3
9	44	Male	1.3	0.5	210	41	36	6.7	3.5	1.1
10	29	Female	0.7	0.2	158	24	19	7.5	4.0	1.0
11	67	Male	8.2	4.0	390	120	165	6.2	2.1	0.6
12	41	Male	1.1	0.3	185	32	29	7.0	3.7	1.2
13	48	Male	0.7	0.2	208	15	30	4.6	2.1	0.8
14	60	Male	22.8	12.6	962	53	41	6.9	3.3	0.9
15	27	Male	1.0	0.3	180	56	111	6.8	3.9	1.85
16	28	Female	0.8	0.2	309	55	23	6.8	4.1	1.51

B.9 Audit Trace: Fold 9 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	51	Male	0.8	0.2	160	34	20	6.9	3.7	1.1
2	41	Female	0.9	0.2	201	31	24	7.6	3.8	1.0
3	55	Male	0.7	0.2	290	53	58	6.8	3.4	1.0
4	48	Female	1.1	0.7	527	178	250	8.0	4.2	1.1
5	50	Male	0.7	0.2	192	18	15	7.4	4.2	1.3
6	49	Male	1.0	0.3	230	48	58	8.4	4.2	1.0
7	44	Male	1.2	0.4	210	36	45	6.7	3.5	1.1
8	53	Female	0.8	0.2	185	28	32	7.1	4.0	1.2
9	38	Male	2.5	1.1	320	92	110	5.8	2.6	0.8
10	61	Male	0.9	0.3	198	41	38	6.3	3.2	1.0
11	47	Female	1.0	0.3	240	55	60	7.5	4.2	1.3
12	35	Male	0.7	0.2	170	22	27	6.9	3.7	1.2
13	26	Male	2.0	0.9	195	24	65	7.8	4.3	1.2
14	47	Male	2.7	1.3	275	123	73	6.2	3.3	1.1
15	74	Male	1.0	0.3	175	30	32	6.4	3.4	1.1
16	48	Male	3.2	1.6	257	33	116	5.7	2.2	0.62

B.10 Audit Trace: Fold 10 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	65	Female	0.7	0.2	182	23	28	6.8	2.9	0.7
2	30	Female	0.7	0.2	63	31	27	5.8	3.4	1.4
3	50	Male	7.3	3.7	92	44	236	6.8	1.6	0.3
4	36	Male	2.8	1.5	305	28	76	5.9	2.5	0.7
5	26	Male	42.8	19.7	390	75	138	7.5	2.6	0.5
6	58	Male	0.8	0.2	298	33	59	6.2	3.1	1.0
7	54	Male	3.2	1.5	312	41	85	6.2	2.3	0.7
8	28	Female	0.8	0.2	175	18	16	7.3	3.8	1.1
9	46	Male	1.1	0.4	210	36	44	6.7	3.2	0.9
10	63	Male	9.4	4.1	420	58	110	5.8	1.9	0.5
11	39	Female	0.7	0.2	198	14	13	7.0	3.0	0.8
12	57	Male	2.6	1.2	267	29	60	6.3	2.7	0.8
13	54	Female	5.5	3.2	350	67	42	7.0	3.2	0.8
14	43	Male	22.5	11.8	143	22	143	6.6	2.1	0.46
15	57	Male	4.5	2.3	315	120	105	7.0	4.0	1.3
16	65	Male	0.7	0.2	199	19	22	6.3	3.6	1.3